Talia Baddour

(310) 503-1899 | tbaddour@ucsd.edu | www.linkedin.com/in/talia-baddour

Professional Overview

I am a bioengineer with mechanical engineering training and six years of research experience spanning computational modeling, biomaterials engineering, and single-cell and spatial genomics. My expertise spans **cardiovascular biology, fibrosis and calcification, epigenetics, and sex differences in disease, and I specialize in generating and analyzing single-cell and spatial transcriptomic data from human tissue to uncover sex-specific disease mechanisms.** I bridge tissue-level omics approaches with 3D hydrogel disease models to uncover clinically relevant mechanisms of heart disease. I am passionate about translating multi-omic insights into precision therapeutics.

Research and Professional Experience

Graduate Student Researcher 2021 – Present

University of California, San Diego, Dept. of Bioengineering, Aguado iBiomaterials Lab

- Developed an integrated spatial and single-cell atlas of immune and fibroblast populations in calcific aortic valves, uncovering sex-specific disease markers such as male-biased *COMP* localization near calcification sites and a female-specific SPP1–CD44 macrophage–fibroblast signaling axis linked to tissue fibrosis (*ATVB*, 2026).
- Independently led end-to-end 10x Genomics single-cell and Visium spatial workflows, from human valve tissue sourcing, processing, and cell isolation to generating omics libraries for sequencing on Illumina systems.
- Built R and Python analysis pipelines to identify sex-specific drivers of calcific aortic valve disease and cardiac fibrosis, including cross-species integration of human and porcine datasets using Seurat and Scanpy workflows (*Science Advances*, 2025; *Advanced Healthcare Materials*, in revision).
- Engineered 3D poly(ethylene glycol) hydrogel platforms (thiol-ene photocrosslinking, MMP-degradable crosslinkers, embedded microparticles) to model human valve disease in vitro. Characterized hydrogel mechanics by rheology and atomic force microscopy.
- Quantified sex differences in valve collagen architecture using polarized light and second harmonic generation microscopy and developed Fiji/ImageJ image analysis workflows. Discovered that sex-specific collagen alignment in aortic valves is associated with female fibroblast activation, which was validated in an in vitro aligned collagen matrix platform (*Matrix Biology*, in revision).
- Mentored 6 trainees (4 undergraduate, 2 master's) in experimental design, laboratory techniques, and data communication.

Post-baccalaureate Researcher, NIH PREP Scholar 2020 – 2021

Virginia Polytechnic Institute and State University, Blacksburg, VA

- Developed an experimentally validated Hill-type muscle–tendon model of murine plantarflexion in MATLAB that accurately predicted Achilles tendon force and stiffness against experimental data (Orthopedic Mechanobiology Lab, PI: Dr. Vincent Wang).
- Built a finite element model of vaginal tissue inflation (FEBio, digital image correlation) to support reduced order modeling of vaginal tearing during childbirth (STRETCH Lab, PI: Dr. Raffaella De Vita).

Undergraduate Researcher 2019 – 2020

Lafayette College, Easton, PA

- Developed a finite element model of convection-enhanced drug delivery to the brain and a MATLAB optimization routine to characterize soft-tissue properties from experimental data. The project was completed as an honors thesis.

Relevant Skills

- **Bioinformatics:** Extensive experience with single-cell RNA-sequencing and Visium spatial transcriptomics analysis, cross-species integration, differential expression, pseudotime, cell–cell communication, GO enrichment, and Ingenuity Pathway Analysis. Proficient in R (Seurat, SoupX, monocle3, biomaRt), Python, MATLAB, Unix/Linux, Bash scripting, cloud computing, CellRanger, and SpaceRanger.
- **Library Preparation:** Highly skilled in hands-on, independent generation of single-cell and Visium spatial transcriptomic libraries (10x Genomics) for next generation sequencing on Illumina systems.
- **Tissue Processing:** Advanced expertise in isolation of cells and tissue from human specimens for multi-omics studies; tissue processing for histology, immunofluorescence, and RNA-FISH. Skilled in primary porcine valve cell isolation and culture.
- **Molecular biology:** Extensive experience in RNA isolation and quality assessment (Agilent TapeStation), PCR primer design, RT-qPCR assay design and validation, siRNA-mediated gene knockdown, and Western blotting.

- **Biomaterials:** Skilled in design of 3D PEG hydrogel models of cardiovascular disease incorporating interpenetrating collagen networks, hydroxyapatite particles, and polystyrene microparticle with thiol-ene photochemistry. Proficient in rheological characterization, and finite element modeling of hydrogel materials.
- **Microscopy and Image Analysis:** Skilled in spinning disk confocal, second harmonic generation, and polarized light microscopy to probe human tissue architecture. Highly skilled in 3D immunocytochemistry analysis using IMARIS.
- **Clinical Cohort & Data Integration:** Experienced in curating human patient sample cohorts and clinical metadata, and in synthesizing clinical and research evidence to identify molecular mediators of heart disease.
- **Mechanistic Disease Modeling:** Expert in engineering in vitro models that recapitulate human disease phenotypes, validated through multi-omic comparison to patient tissue, and applying them to define genetic mechanisms of disease, with specialized knowledge of fibroblast activation (myofibroblast, osteoblast-like) and immune cell phenotypes in heart disease.

Accomplishments and Presentations

- 2026, Poster Presentation, EMBL Symposium: Sex Differences in Health and Disease
- 2026, Awardee, DC Advocacy Travel Grant, UC Graduate and Professional Council
- 2025, Oral Presentation, BMES Annual Meeting
- 2025, Poster Presentation, Tissue Engineering and Regenerative Medicine International Society Conference
- 2023, Fellow, NHLBI T32 Graduate Research Training Grant; San Diego Match Fellowship
- 2023, Poster Presentation, Gordon Research Conference: Sex Differences in Immunity;
- 2022, Oral Presentation, BMES Annual Meeting
- 2022, Awardee, Travel Award to AIMBE Public Policy Institute
- 2021, Poster Presentation, Orthopedic Research Society Annual Meeting
- 2020, Awardee, Mechanical Engineering Department Award, Lafayette College

Teaching, Mentorship, and Leadership

- 2026, Student Advocate, UC Graduate and Professional Council. Compiled testimony from 10 graduate students across 4 departments into a report on federal funding cuts and met with Senator Adam Schiff's office in Washington, DC
- 2021 – 2024, Chair of Professional Development, Gender Minorities in Bioengineering (GMBE) at UCSD
- 2021 – 2024, Founder, GMBE Networking Night. Connected 150 bioengineering students with San Diego start-ups and biotech companies; 70% of attendees reported a resulting career connection
- 2021 – Present, Research mentor to 4 undergraduate and 2 master's students; undergraduate mentee received the Faculty Choice Poster Award at the UCSD Undergraduate Engineering Research Symposium
- 2022 – 2023, Teaching Assistant, UCSD BENG 160 Chemical and Molecular Engineering Techniques (70+ students) and BENG 207 Biomaterials for Medical Device Design (20+ students)

Peer-reviewed Publications

- **Baddour, T.** et al. Spatial and single-cell mapping reveals valvular interstitial cell and macrophage sex differences in calcific aortic valve disease. *Arteriosclerosis Thrombosis Vascular Biology*. 2026;46(7):e324694. doi:10.1161/ATVBAHA.126.324694
- Gorashi, R.M., **Baddour, T.** et al. Y chromosome-linked UTY modulates sex differences in valvular fibroblast methylation in response to nanoscale extracellular matrix cues. *Science Advances*. 2025;11(11):eads5717. doi:10.1126/sciadv.ads5717
- Gorashi, R.M., **Baddour, T.** et al. Sex chromosome and gonadal regulation of cardiac myofibroblast activation decoupled using mechanically tunable hydrogels and the Four Core Genotypes model. *Advanced Healthcare Materials* (in revision)
- Wennerholm, K., **Baddour, T**. et al. Sex-associated valvular interstitial cell responses to an alignment tunable collagen hydrogel platform. *Matrix Biology* (in revision)

Education

University of California, San Diego, CA
PhD in Bioengineering, expected 03/2027
GPA: 3.74

Lafayette College, Easton, PA
B.S. in Mechanical Engineering, 05/2020
GPA: 3.82